

Stakeholder Engagement

Team Gambia — TRI AI Saturdays Cohort 10 — Agricultural Extension RAG: Smart Retrieval for Farmers

Stakeholder Map

Smallholder farmers	Primary end beneficiaries of any downstream system built on this retrieval component; directly affected by the system's performance.
Agricultural extension officers	"Plant doctors" (e.g. under Uganda's MAAIF) who currently perform the diagnosis role this system aims to replace.
Regional agricultural research organizations	Organizations such as NARO/Abi ZARDI, CGIAR, FAO, and ICRISAT, whose factsheets and research reports form the core of the dataset.
Competition organizers (TRI AI)	Defined the dataset, evaluation metric, and task scope; a stakeholder in ensuring submitted work meets the challenge's goals.
Cohort mentors and TRI AI Saturdays program	Provided guidance and evaluation criteria for the broader learning objectives of the cohort.
Future developers of agricultural RAG tools	Stakeholders who may build on or learn from this project's documented findings and approach.

Engagement Process

Given the project's scope as a cohort-based, time-boxed technical challenge built on a pre-defined Kaggle competition dataset, direct field engagement with farmers or extension officers was outside the project's scope and timeline. Stakeholder needs were instead incorporated through the following channels:

- Domain-grounded query and terminology mapping:** we researched documented regional agricultural terminology (e.g. local names for Banana Bacterial Wilt, Bean Leaf Beetle, Cassava Brown Streak Disease relevant to the West Nile Agro-Ecological Zone of Uganda) and incorporated a query-expansion mechanism to bridge the gap between how farmers are documented to phrase problems and the scientific vocabulary of the extension corpus.
- Cohort mentor feedback:** iterative guidance from TRI AI Saturdays mentors shaped both the technical approach and the framing of this project's documentation and evaluation methodology.
- Dataset design as an implicit stakeholder signal:** the competition dataset's deliberate inclusion of hard negatives and graded relevance reflects the organizers' (TRI AI's) understanding of what matters to end stakeholders — namely, that a wrong-but-plausible answer is worse than an unhelpful one, which directly shaped our evaluation priorities (per-query error analysis over aggregate metrics alone).

What We Would Do With More Time or Access

- Direct interviews or structured surveys with smallholder farmers and extension officers in the target region, to validate whether the query phrasings in the training set and our expansion dictionary genuinely reflect how real farmers ask questions.
- Field validation of retrieval outputs with an agronomist or extension officer, to catch cases where a retrieved document is lexically/semantically plausible but agronomically inappropriate in ways a

relevance label alone would not capture.

- Engagement with regional agricultural research institutions (e.g. NARO/Abi ZARDI) to expand and validate the local-terminology mapping used in query expansion, beyond the six crop/pest patterns currently covered.

Limitations of This Engagement

This project's stakeholder engagement was necessarily indirect and literature/dataset-mediated rather than field-based, given the cohort's time constraints and the competition's fixed dataset scope. We flag this explicitly as a limitation: the system's retrieval quality has been validated against expert-constructed relevance labels, not against real-world usage or feedback from the farmers and extension officers it is ultimately intended to serve.